

Project 03 — IT BA Project

Software Feature Specification

Feature: Intelligent Order Modification & Cancellation System

Company: UrbanThread (Simulated)

Project Type: Business Analysis + IT Business Analysis

This is a student portfolio project. All company details, metrics, stakeholder interviews, user data, and business context are simulated / fictional unless explicitly stated otherwise. No real client work or production data is represented.

1. Executive Summary

This project presents a comprehensive Business Analysis case study for a proposed software feature: an Intelligent Order Modification & Cancellation System for a fictional e-commerce company, UrbanThread (simulated). The current manual, fragmented process creates operational inefficiencies, customer dissatisfaction, and revenue leakage. The proposed solution is a self-service, rules-based order modification and cancellation feature integrated into the customer account portal and mobile app, enabling customers to modify or cancel eligible orders within defined business rules, with automated inventory release, payment adjustments, and notifications.

Key Deliverables: Business Requirements (12), Functional Requirements (18), Non-Functional Requirements (10), User Stories (24), Acceptance Criteria (35+), Use Cases (8), Traceability Matrix, UAT Scenarios (30), Risk Register, KPI Framework.

2. Company / Product Context

2.1 Company Profile (Simulated)

Attribute	Details
Company Name	UrbanThread (fictional)
Industry	E-commerce — Fashion & Apparel
Founded	2018 (simulated)
Headquarters	Austin, Texas (simulated)
Business Model	B2C online retail with direct-to-consumer fulfillment
Product	Web and mobile e-commerce platform selling apparel, accessories, footwear
Target Customers	Adults aged 22–45, fashion-conscious, mobile-first shoppers
Annual Revenue	~\$45M (simulated)
Annual Orders	~850,000 (simulated)
Average Order Value	~\$53 (simulated)
Customer Service Team	45 agents (simulated)
Warehouse Partners	2 third-party fulfillment centers (simulated)
Payment Processor	Stripe (simulated integration)
Shipping Partners	FedEx, UPS, USPS (simulated integrations)

2.2 Digital Ecosystem (Simulated)

System	Purpose	Integration Type
Customer Web Portal	Order placement, account management	Internal
Mobile App (iOS/Android)	Order placement, account management	Internal
Order Management System (OMS)	Order lifecycle management	Internal
Warehouse Management System (WMS)	Fulfillment operations	API
Payment Gateway	Payment processing, refunds	Stripe API
Shipping Aggregator	Label generation, tracking	API
CRM / Support Desk	Customer service ticketing	Zendesk (simulated)
Notification Service	Email, SMS, push notifications	Internal + Twilio/SendGrid

3. Business Problem

UrbanThread's customer service team is overwhelmed by order modification and cancellation requests that could be handled through self-service. The current process is manual, error-prone, and inconsistent, leading to: high operational cost, customer friction, revenue leakage, inventory inefficiency, inconsistent outcomes, and limited scalability.

3.1 Problem Quantification (Simulated)

Metric	Value	Source
Monthly orders	~70,000	Simulated OMS data
Monthly modification/cancellation requests	~2,450	Simulated CRM data
% handled via support	100%	Current state
Average handling time	10 minutes	Simulated
Monthly support hours consumed	~408 hours	Calculated
Estimated monthly support cost	~\$18,000	Simulated
CSAT for these interactions	3.2/5	Simulated
Orders shipped despite cancellation request	~8%	Simulated

4. Problem Statement

UrbanThread customers cannot independently modify or cancel orders after placement, forcing them to contact customer support. This creates operational burden, inconsistent customer experiences, delayed resolutions, and unnecessary shipping costs. There is no self-service capability that applies consistent business rules, validates eligibility in real-time, and automates downstream actions such as inventory release, payment adjustment, and notifications.

5. Business Objective

Primary: Enable customers to self-serve order modifications and cancellations within defined business rules, reducing support contacts and improving customer satisfaction while maintaining operational control.

5.1 Secondary Objectives

ID	Objective	Measurable Outcome (Proposed Target)
OBJ-001	Reduce support contacts for order changes	40% reduction within 6 months of launch
OBJ-002	Improve customer satisfaction for order change experience	CSAT ≥ 4.2/5 within 3 months

ID	Objective	Measurable Outcome (Proposed Target)
OBJ-003	Reduce unnecessary shipments due to late cancellations	50% reduction in shipped-then-returned orders
OBJ-004	Improve operational efficiency	Reduce handling time for eligible requests to < 2 minutes
OBJ-005	Increase customer self-service adoption	60% of eligible requests handled via self-service
OBJ-006	Reduce revenue leakage from order abandonment	15% reduction in cart/order abandonment post-change request

6. Scope

6.1 In Scope

Area	Description
Order modification	Change delivery address, delivery speed, payment method (for eligible orders)
Order cancellation	Full order cancellation for eligible orders
Item-level cancellation	Cancel individual items within a multi-item order
Eligibility engine	Real-time rules evaluation for modification/cancellation
Inventory release	Automatic inventory reallocation upon cancellation
Payment adjustment	Automatic refund or payment method update
Notifications	Email, SMS, push notifications for status changes
Audit trail	Logging of all modification/cancellation actions
Customer portal UI	Web and mobile interfaces for self-service
Support agent view	Read-only visibility of customer actions for context
Reporting	Operational dashboards for modification/cancellation metrics

6.2 Out of Scope

Area	Reason
Returns and exchanges	Separate feature initiative
Post-delivery modifications	Different business process
Subscription orders	Not applicable to current model
B2B/wholesale orders	Different rules and workflows
International orders	Phase 2 consideration
Partial quantity modification	Phase 2
Split shipment modifications	Complexity deferred
Real-time carrier interception	Not technically feasible with current partners

7. Stakeholder Analysis

ID	Stakeholder	Role	Interest	Influence	Strategy
STK-001	VP of Customer Experience	Executive Sponsor	High	High	Manage closely; weekly updates
STK-002	Head of Customer Support	Business Owner	High	High	Collaborate; requirements validation
STK-003	Product Manager	Product Owner	High	High	Daily collaboration; backlog prioritization
STK-004	Engineering Lead	Technical Lead	Medium	High	Consult on feasibility; technical review
STK-005	UX Designer	Design Lead	High	Medium	Collaborate on wireframes; usability
STK-006	QA Lead	Testing Lead	Medium	Medium	Consult on test scenarios
STK-007	Finance Manager	Payment/Refund Stakeholder	Medium	Medium	Consult on refund rules
STK-008	Warehouse Operations Manager	Fulfillment Stakeholder	High	Medium	Consult on inventory release timing
STK-009	Customer Support Agents	End Users (Internal)	High	Low	Inform; gather feedback
STK-010	Customers	End Users (External)	High	Low	Represent via personas; usability testing
STK-011	Legal/Compliance	Compliance	Low	Medium	Consult on policy requirements
STK-012	Data/Analytics Team	Reporting	Medium	Low	Consult on KPI tracking

8. Current-State / AS-IS Analysis

The current order modification and cancellation process is entirely manual and support-dependent. The customer contacts support; an agent manually reviews the order and checks eligibility; the agent updates OMS; OMS pushes to WMS; the agent processes refunds; and the agent notifies the customer. See the AS-IS Process Flow diagram for the swimlane view.

8.1 AS-IS Pain Points

ID	Pain Point	Impact	Affected Stakeholder
PP-001	Customers must contact support for any order change	High friction; poor experience	Customer
PP-002	No real-time eligibility visibility	Customers wait for agent response	Customer
PP-003	Manual eligibility checking	Inconsistent decisions; errors	Support Agent
PP-004	Average 10-minute handling time per request	High operational cost	Support Manager
PP-005	Agents use multiple systems (CRM, OMS, WMS, payment)	Inefficiency; error-prone	Support Agent
PP-006	No automated inventory release	Items shipped unnecessarily	Warehouse
PP-007	Refund processing delays	Customer dissatisfaction	Customer, Finance
PP-008	No audit trail of who changed what	Compliance risk; no accountability	Legal/Compliance
PP-009	Peak volume overwhelms support	Long wait times; abandoned requests	Support Manager
PP-010	Inconsistent communication	Customer confusion	Customer

8.2 AS-IS Bottlenecks and Failure Points

ID	Bottleneck / Failure Point	Description
BF-001	Support queue wait time	Average 15-minute wait during peak hours

ID	Bottleneck / Failure Point	Description
BF-002	Manual eligibility check	Agent must check OMS, WMS, and shipping status manually
BF-003	OMS update lag	WMS may not receive update before picking begins
BF-004	Refund processing	3–5 business days for refund to appear
BF-005	No self-service fallback	If support is unavailable, customer cannot act
BF-006	Escalation delays	Senior agent approval adds 24+ hours

9. Root Cause Analysis

9.1 5 Whys

#	Question	Answer
1	Why do customers contact support for order changes?	Because there is no self-service capability in the portal or app.
2	Why is there no self-service capability?	Because the business rules for eligibility are complex and not codified.
3	Why are the rules not codified?	Because eligibility depends on real-time data from multiple systems (OMS, WMS, payment, shipping) that are not integrated for this purpose.
4	Why are the systems not integrated?	Because the feature was never prioritized; manual process was "good enough" at lower volumes.
5	Why was it never prioritized?	Because the business impact was not quantified; support team absorbed the workload without formal escalation.

Root Cause: Lack of a codified, real-time eligibility engine and integrated self-service interface, stemming from under-prioritization due to unquantified business impact.

9.2 Root Causes → Proposed Feature

Root Cause	Proposed Feature Response
No self-service UI	Customer portal and mobile app interfaces for modification/cancellation
Complex, uncoded rules	Rules engine that codifies eligibility criteria
Systems not integrated	API layer that orchestrates OMS, WMS, payment, shipping
Manual checks	Automated real-time eligibility evaluation
Inconsistent decisions	Standardized rules applied uniformly
No audit trail	Comprehensive logging of all actions
Delayed refunds	Automated payment adjustment triggers
Inventory not released	Automated inventory release upon cancellation

10. Proposed Solution

The Intelligent Order Modification & Cancellation System is a self-service feature that enables customers to modify or cancel eligible orders directly from their account portal or mobile app.

10.1 Solution Components

Component	Description
Customer Portal UI	Web interface for order modification/cancellation
Mobile App UI	iOS/Android interface for order modification/cancellation
Eligibility Rules Engine	Backend service that evaluates order eligibility

Component	Description
Order Orchestration Service	Coordinates actions across OMS, WMS, payment, shipping
Payment Adjustment Service	Handles refunds and payment method updates
Inventory Release Service	Triggers inventory reallocation in WMS
Notification Service	Sends email, SMS, push notifications
Audit Log Service	Records all actions for compliance and reporting
Support Agent Dashboard	Read-only view of customer actions for context
Reporting Dashboard	Operational metrics for management

10.2 Key Design Principles

Principle	Description
Customer-first	Minimize friction; clear, immediate feedback
Rules-based	Consistent, auditable decisions
Real-time	Eligibility evaluated at the moment of request
Resilient	Graceful handling of system failures
Transparent	Clear communication of what can and cannot be changed
Secure	Authentication and authorization for all actions
Scalable	Handles peak volumes without degradation

11. Future-State / TO-BE Process

The future-state process enables customers to self-serve order modifications and cancellations through the portal or app, with automated backend processing and clear communication. See the TO-BE Process Flow diagram.

11.1 What Changes vs. What Stays the Same

Aspect	AS-IS	TO-BE	Change
Customer action	Contact support	Self-service portal/app	Changed
Eligibility check	Manual by agent	Automated rules engine	Changed
Decision consistency	Agent-dependent	Standardized rules	Changed
Processing time	8–12 minutes	< 2 minutes	Changed
Inventory release	Manual / delayed	Automated	Changed
Refund processing	Manual trigger	Automated trigger	Changed
Notifications	Agent-initiated	Automated	Changed
Audit trail	Minimal	Comprehensive	Changed
Support role	Handle all requests	Handle exceptions only	Changed
Order data	Same OMS	Same OMS	Unchanged
Payment gateway	Stripe	Stripe	Unchanged
Shipping partners	FedEx, UPS, USPS	Same	Unchanged
Warehouse operations	Same WMS	Same WMS	Unchanged

12. Business Requirements

ID	Requirement	Rationale	Priority	Source
BR-001	The system shall enable customers to self-serve order modifications and cancellations without contacting support.	Reduces support burden; improves customer experience.	Must Have	STK-001, STK-002
BR-002	The system shall apply consistent, codified business rules to determine order modification and cancellation eligibility.	Ensures fair and consistent customer treatment; reduces errors.	Must Have	STK-002
BR-003	The system shall process eligible modification and cancellation requests in real-time (within 2 minutes).	Improves customer satisfaction; reduces wait times.	Must Have	STK-001
BR-004	The system shall automatically release inventory for cancelled items to enable reallocation.	Reduces unnecessary shipping costs; improves inventory efficiency.	Must Have	STK-008
BR-005	The system shall automatically process payment adjustments (refunds or payment method updates) for eligible changes.	Reduces manual finance work; improves customer trust.	Must Have	STK-007
BR-006	The system shall send automated notifications to customers at each stage of the modification/cancellation process.	Keeps customers informed; reduces follow-up contacts.	Must Have	STK-001
BR-007	The system shall maintain a comprehensive audit trail of all modification and cancellation actions.	Supports compliance; enables dispute resolution.	Should Have	STK-011
BR-008	The system shall provide support agents with visibility into customer self-service actions for context.	Enables effective exception handling; improves support efficiency.	Should Have	STK-002
BR-009	The system shall provide operational reporting on modification/cancellation volumes, success rates, and processing times.	Enables continuous improvement; supports KPI tracking.	Should Have	STK-001, STK-012

ID	Requirement	Rationale	Priority	Source
BR-010	The system shall handle peak volumes (up to 3x normal) without degradation in performance.	Ensures reliability during high-demand periods.	Should Have	STK-004
BR-011	The system shall comply with applicable data protection and payment card industry (PCI) requirements.	Legal and regulatory compliance.	Must Have	STK-011
BR-012	The system shall reduce support contacts related to order changes by at least 40% within 6 months of launch.	Measurable business outcome; justifies investment.	Must Have	STK-001

13. Functional Requirements

ID	Requirement	Rationale	Priority	Source	Traces To
FR-001	The system shall display a "Modify Order" and "Cancel Order" option on the order details page for eligible orders.	Provides entry point for self-service.	Must Have	STK-001	BR-001
FR-002	The system shall evaluate order eligibility for modification and cancellation in real-time based on order status, fulfillment stage, and business rules.	Ensures only eligible orders can be changed.	Must Have	STK-002	BR-002
FR-003	The system shall display the specific reason when an order is not eligible for modification or cancellation.	Provides transparency; reduces confusion.	Must Have	STK-001	BR-001
FR-004	The system shall allow customers to modify the delivery address for eligible orders.	Addresses common modification need.	Must Have	STK-001	BR-001
FR-005	The system shall allow customers to change the shipping speed (standard to express or express to standard) for eligible orders.	Addresses common modification need; potential revenue impact.	Should Have	STK-001	BR-001
FR-006	The system shall allow customers to cancel individual items within a multi-item order if eligible.	Provides flexibility; reduces full-order cancellations.	Should Have	STK-001	BR-001
FR-007	The system shall allow customers to cancel an entire order if eligible.	Core self-service capability.	Must Have	STK-001	BR-001
FR-008	The system shall calculate and display any applicable fees, refunds, or payment adjustments before the customer confirms the action.	Transparency; reduces disputes.	Must Have	STK-007	BR-005
FR-009	The system shall require customer confirmation before processing any modification or cancellation.	Prevents accidental changes.	Must Have	STK-001	BR-001
FR-010	The system shall update the Order Management System (OMS) with the modification or cancellation in real-time.	Ensures order data accuracy.	Must Have	STK-004	BR-002
FR-011	The system shall trigger inventory release in the Warehouse Management System (WMS) for cancelled items.	Enables inventory reallocation.	Must Have	STK-008	BR-004
FR-012	The system shall trigger payment adjustment (refund or payment method update) through the payment gateway.	Automates financial reconciliation.	Must Have	STK-007	BR-005
FR-013	The system shall send an email confirmation to the customer upon successful modification or cancellation.	Keeps customer informed.	Must Have	STK-001	BR-006
FR-014	The system shall send an SMS notification to the customer upon successful modification or cancellation (if opted in).	Provides real-time updates.	Should Have	STK-001	BR-006
FR-015	The system shall send a push notification to the customer upon successful modification or cancellation (if app installed).	Provides real-time updates.	Should Have	STK-001	BR-006
FR-016	The system shall log all modification and cancellation actions with timestamp, user ID, action type, and before/after values.	Supports audit and compliance.	Should Have	STK-011	BR-007
FR-017	The system shall provide a read-only view of customer self-service actions in the support agent dashboard.	Enables context for exception handling.	Should Have	STK-002	BR-008
FR-018	The system shall generate operational reports on modification/cancellation metrics (volume, success rate, processing time, reasons).	Supports continuous improvement.	Should Have	STK-012	BR-009

14. Non-Functional Requirements

ID	Requirement	Rationale	Priority	Source	Traces To
NFR-001	The system shall evaluate eligibility and display options within 3 seconds of the customer initiating a request (95th percentile).	Ensures responsive experience.	Must Have	STK-004	BR-003
NFR-002	The system shall process a confirmed modification or cancellation within 60 seconds (95th percentile).	Ensures timely completion.	Must Have	STK-004	BR-003
NFR-003	The system shall support up to 500 concurrent modification/cancellation requests without degradation.	Handles peak volumes.	Should Have	STK-004	BR-010
NFR-004	The system shall be available 99.9% of the time (excluding planned maintenance).	Ensures reliability.	Must Have	STK-004	BR-001
NFR-005	The system shall encrypt all data in transit (TLS 1.2+) and at rest (AES-256).	Security compliance.	Must Have	STK-011	BR-011
NFR-006	The system shall comply with PCI DSS requirements for payment data handling.	Regulatory compliance.	Must Have	STK-011	BR-011
NFR-007	The system shall be accessible via web (Chrome, Firefox, Safari, Edge) and mobile (iOS 14+, Android 10+).	Broad accessibility.	Must Have	STK-005	BR-001
NFR-008	The system shall meet WCAG 2.1 AA accessibility standards.	Inclusive design.	Should Have	STK-005	BR-001
NFR-009	The system shall provide clear, actionable error messages to users in plain language.	Usability; reduces frustration.	Must Have	STK-005	BR-001
NFR-010	The system shall log all errors and exceptions for debugging and monitoring.	Operational visibility.	Should Have	STK-004	BR-009

15. User Stories

ID	User Story	Priority	Traces To
US-001	As a customer, I want to see if my order is eligible for modification or cancellation, so that I know my options without contacting support.	Must Have	FR-001, FR-002
US-002	As a customer, I want to understand why my order is not eligible for modification or cancellation, so that I understand the limitation.	Must Have	FR-003
US-003	As a customer, I want to change my delivery address on an eligible order, so that my order goes to the correct location.	Must Have	FR-004
US-004	As a customer, I want to change my shipping speed on an eligible order, so that I can receive my order faster or slower.	Should Have	FR-005
US-005	As a customer, I want to cancel individual items in my order, so that I can remove items I no longer want.	Should Have	FR-006
US-006	As a customer, I want to cancel my entire order, so that I can get a full refund.	Must Have	FR-007
US-007	As a customer, I want to see any fees or refunds before confirming my change, so that I understand the financial impact.	Must Have	FR-008
US-008	As a customer, I want to confirm my change before it is processed, so that I don't make accidental changes.	Must Have	FR-009
US-009	As a customer, I want to receive a confirmation email after my change is processed, so that I have a record of the change.	Must Have	FR-013
US-010	As a customer, I want to receive an SMS notification after my change is processed, so that I know immediately.	Should Have	FR-014
US-011	As a customer, I want to receive a push notification after my change is processed, so that I know immediately.	Should Have	FR-015
US-012	As a customer, I want to see the updated order status after my change, so that I know the change was successful.	Must Have	FR-010
US-013	As a support agent, I want to see what self-service actions a customer has taken, so that I can provide informed support.	Should Have	FR-017
US-014	As a support agent, I want to view the audit trail of changes, so that I can resolve disputes.	Should Have	FR-016
US-015	As a support agent, I want to handle exception cases that cannot be self-served, so that I can resolve complex issues.	Must Have	FR-003
US-016	As a support manager, I want to see reports on modification/cancellation volumes and success rates, so that I can monitor performance.	Should Have	FR-018
US-017	As a support manager, I want to see the reasons for ineligibility, so that I can identify process improvements.	Could Have	FR-003, FR-018
US-018	As a finance manager, I want to see refund processing reports, so that I can reconcile payments.	Should Have	FR-012, FR-018
US-019	As a warehouse manager, I want to see inventory release reports, so that I can plan operations.	Should Have	FR-011, FR-018
US-020	As a compliance officer, I want to access audit logs, so that I can ensure regulatory compliance.	Should Have	FR-016
US-021	As a system, I want to validate order eligibility in real-time, so that only eligible orders are modified.	Must Have	FR-002
US-022	As a system, I want to handle concurrent modification requests gracefully, so that data integrity is maintained.	Must Have	FR-002, FR-010
US-023	As a system, I want to retry failed payment adjustments, so that refunds are processed reliably.	Should Have	FR-012
US-024	As a system, I want to notify administrators of integration failures, so that issues are resolved quickly.	Should Have	FR-011, FR-012

16. Acceptance Criteria

ID	User Story	Criteria	Type
AC-001	US-001	Given a customer is logged in and views an order that is eligible for modification, When the order details page loads, Then the system shall display "Modify Order" and "Cancel Order" buttons.	Positive
AC-002	US-001	Given a customer is logged in and views an order that is not eligible for modification, When the order details page loads, Then the system shall display a message stating "This order can no longer be modified or cancelled."	Negative
AC-003	US-001	Given a customer is logged in and views an order that is partially eligible, When the order details page loads, Then the system shall display "Cancel Order" only for eligible items.	Boundary
AC-004	US-002	Given an order is not eligible for cancellation, When the customer selects "Cancel Order", Then the system shall display the reason: "This order has already been shipped and cannot be cancelled."	Positive
AC-005	US-002	Given an order is not eligible for modification, When the customer attempts to modify the delivery address, Then the system shall prevent the action and display: "Delivery address cannot be changed once the order is being prepared for shipment."	Negative
AC-006	US-003	Given an order is eligible for address modification, When the customer enters a new valid address and confirms, Then the system shall update the delivery address in the OMS and display a success message.	Positive
AC-007	US-003	Given an order is eligible for address modification, When the customer enters an invalid address, Then the system shall display a validation error and prevent submission.	Negative
AC-008	US-003	Given an order is eligible for address modification, When the customer enters an address outside the serviceable area, Then the system shall display: "We do not currently deliver to this address."	Boundary
AC-009	US-006	Given an order is eligible for cancellation, When the customer confirms cancellation, Then the system shall cancel the order, initiate a refund, release inventory, and send a confirmation email.	Positive
AC-010	US-006	Given an order is not eligible for cancellation, When the customer attempts to cancel, Then the system shall display the reason and prevent cancellation.	Negative
AC-011	US-006	Given an order is eligible for cancellation and the customer has already used a discount code, When the customer cancels the order, Then the system shall cancel the order and the discount code shall not be reissued (per policy).	Boundary
AC-012	US-007	Given an order is eligible for cancellation, When the customer selects "Cancel Order", Then the system shall display the refund amount, including any deductions for shipping or processing fees.	Positive
AC-013	US-007	Given an order is eligible for cancellation but the refund amount is \$0, When the customer selects "Cancel Order", Then the system shall display: "This item is non-refundable. Cancellation will not result in a refund."	Negative
AC-014	US-008	Given the customer has selected a modification action, When the customer clicks "Confirm Change", Then the system shall display a confirmation dialog summarizing the change.	Positive
AC-015	US-008	Given the customer has selected a modification action, When the customer clicks "Cancel" in the confirmation dialog, Then the system shall return to the order details page without making changes.	Negative
AC-016	US-009	Given a modification or cancellation is successfully processed, When the processing is complete, Then the system shall send an email to the customer's registered email address within 5 minutes.	Positive
AC-017	US-009	Given a modification or cancellation fails, When the failure occurs, Then the system shall send an email to the customer explaining the failure and next steps.	Negative
AC-018	US-012	Given a modification is successfully processed, When the customer views the order details page, Then the system shall display the updated order information (e.g., new address, new shipping speed).	Positive

ID	User Story	Criteria	Type
AC-019	US-021	Given a customer initiates a modification request, When the system evaluates eligibility, Then the system shall check order status, fulfillment stage, and business rules in real-time.	Positive
AC-020	US-021	Given the eligibility service is unavailable, When the customer initiates a modification request, Then the system shall display: "We are unable to process your request at this time. Please try again later or contact support."	Negative
AC-021	US-022	Given two modification requests are submitted for the same order simultaneously, When the system processes the requests, Then the first request shall be processed and the second shall be rejected with a message: "This order has already been modified. Please refresh and try again."	Positive

17. Use Cases

UC-001 — View Order Eligibility

Field	Details
Actor	Customer
Trigger	Customer navigates to order details page
Preconditions	Customer is logged in; order exists
Main Flow	1. Customer navigates to order details. 2. System retrieves order status. 3. System evaluates eligibility. 4. System displays eligibility status and available actions.
Alternative Flow	3a. Order is not eligible: System displays reason.
Exception Flow	2a. Order not found: System displays error. 3a. Eligibility service unavailable: System displays fallback message.
Postconditions	Customer sees eligibility status and available actions.

UC-002 — Modify Delivery Address

Field	Details
Actor	Customer
Trigger	Customer selects "Modify Address"
Preconditions	Order is eligible for address modification; customer is logged in
Main Flow	1. Customer selects "Modify Address." 2. System displays address form. 3. Customer enters new address. 4. System validates address. 5. Customer confirms. 6. System updates OMS. 7. System sends confirmation.
Alternative Flow	4a. Address invalid: System displays validation error. 5a. Customer cancels: System returns to order details.
Exception Flow	6a. OMS update fails: System displays error and logs issue.
Postconditions	Delivery address updated; customer notified.

UC-003 — Change Shipping Speed

Field	Details
Actor	Customer
Trigger	Customer selects "Change Shipping Speed"
Preconditions	Order is eligible for shipping speed change; customer is logged in
Main Flow	1. Customer selects "Change Shipping Speed." 2. System displays available options with costs. 3. Customer selects new speed. 4. System calculates price difference. 5. Customer confirms. 6. System processes payment adjustment (if applicable). 7. System updates OMS. 8. System sends confirmation.
Alternative Flow	4a. Price difference is \$0: No payment adjustment.
Exception Flow	6a. Payment adjustment fails: System displays error and does not process change.
Postconditions	Shipping speed updated; payment adjusted; customer notified.

UC-004 — Cancel Individual Item

Field	Details
Actor	Customer
Trigger	Customer selects "Cancel" for a specific item
Preconditions	Order contains multiple items; at least one item is eligible for cancellation

Field	Details
Main Flow	1. Customer selects "Cancel" for an item. 2. System validates item eligibility. 3. System displays refund amount for item. 4. Customer confirms. 5. System updates OMS. 6. System releases inventory for item. 7. System processes refund. 8. System sends confirmation.
Alternative Flow	2a. Item not eligible: System displays reason. 4a. Customer cancels: System returns to order details.
Exception Flow	7a. Refund fails: System logs issue and notifies finance team.
Postconditions	Item cancelled; refund processed; inventory released; customer notified.

UC-005 — Cancel Entire Order

Field	Details
Actor	Customer
Trigger	Customer selects "Cancel Order"
Preconditions	Order is eligible for cancellation
Main Flow	1. Customer selects "Cancel Order." 2. System displays refund amount. 3. Customer confirms. 4. System cancels order in OMS. 5. System releases all inventory. 6. System processes refund. 7. System sends confirmation.
Alternative Flow	2a. Refund amount is \$0: System displays message. 3a. Customer cancels: System returns to order details.
Exception Flow	4a. OMS cancellation fails: System displays error. 6a. Refund fails: System logs issue.
Postconditions	Order cancelled; refund processed; inventory released; customer notified.

UC-006 — View Modification History (Support Agent)

Field	Details
Actor	Support Agent
Trigger	Agent opens customer record
Preconditions	Agent is logged in; customer has modification history
Main Flow	1. Agent opens customer record. 2. System displays list of self-service actions. 3. Agent selects an action to view details. 4. System displays audit trail.
Alternative Flow	2a. No actions: System displays "No self-service actions."
Exception Flow	None
Postconditions	Agent has context for support interaction.

UC-007 — Handle Exception Case

Field	Details
Actor	Support Agent
Trigger	Customer contacts support for ineligible request
Preconditions	Agent is logged in; customer has an ineligible request
Main Flow	1. Agent reviews order. 2. Agent explains ineligibility. 3. Agent offers alternatives (e.g., return after delivery). 4. Agent documents interaction.
Alternative Flow	2a. Agent identifies system error: Agent escalates to technical team.
Exception Flow	None
Postconditions	Customer informed; interaction documented.

UC-008 — Generate Operational Report

Field	Details
Actor	Support Manager

Field	Details
Trigger	Manager requests report
Preconditions	Manager is logged in; data is available
Main Flow	1. Manager selects report type. 2. System generates report. 3. Manager views/downloads report.
Alternative Flow	2a. No data: System displays "No data available."
Exception Flow	2a. Report generation fails: System displays error.
Postconditions	Manager has report for analysis.

18. MoSCoW Prioritization

18. Must Have (Phase 1 — MVP)

ID	Requirement	Reasoning
BR-001	Self-service capability	Core value proposition
BR-002	Codified eligibility rules	Ensures consistency
BR-003	Real-time processing	Key differentiator
BR-004	Inventory release	Operational efficiency
BR-005	Payment adjustment	Financial accuracy
BR-006	Automated notifications	Customer experience
BR-011	Compliance	Legal requirement
FR-001	Eligibility display	Entry point
FR-002	Real-time eligibility evaluation	Core functionality
FR-003	Ineligibility reason	Transparency
FR-004	Address modification	High-demand feature
FR-007	Full order cancellation	High-demand feature
FR-008	Fee/refund display	Transparency
FR-009	Confirmation	Error prevention
FR-010	OMS update	Data integrity
FR-011	Inventory release	Operational efficiency
FR-012	Payment adjustment	Financial accuracy
FR-013	Email confirmation	Customer communication
NFR-001	Response time	Usability
NFR-002	Processing time	Usability
NFR-004	Availability	Reliability
NFR-005	Encryption	Security
NFR-006	PCI compliance	Legal
NFR-007	Cross-platform	Accessibility
NFR-009	Clear error messages	Usability

18. Should Have (Phase 2)

ID	Requirement	Reasoning
BR-007	Audit trail	Compliance; dispute resolution
BR-008	Support agent visibility	Exception handling
BR-009	Operational reporting	Continuous improvement
BR-010	Peak volume handling	Scalability
FR-005	Shipping speed change	Additional flexibility
FR-006	Item-level cancellation	Additional flexibility
FR-014	SMS notification	Enhanced communication
FR-015	Push notification	Enhanced communication
FR-016	Audit logging	Compliance
FR-017	Support agent dashboard	Exception handling

ID	Requirement	Reasoning
FR-018	Operational reports	Management visibility
NFR-003	Concurrency support	Scalability
NFR-008	Accessibility	Inclusivity
NFR-010	Error logging	Operational visibility

18. Could Have (Phase 3)

ID	Requirement	Reasoning
—	Partial quantity modification	Lower demand; complex
—	International order support	Phase 2 consideration
—	Real-time carrier interception	Technical limitations
—	AI-powered modification suggestions	Nice-to-have

18. Won't Have (Out of Scope)

ID	Requirement	Reasoning
—	Returns and exchanges	Separate initiative
—	Post-delivery modifications	Different process
—	Subscription orders	Not applicable
—	B2B/wholesale orders	Different rules
—	Split shipment modifications	Complexity deferred

19. Feature / Requirement Traceability Matrix

BR ID	FR ID	US ID	AC ID	UAT ID
BR-001	FR-001	US-001	AC-001, AC-002, AC-003	UAT-001, UAT-002
BR-001	FR-002	US-001	AC-019, AC-020	UAT-003
BR-001	FR-003	US-002	AC-004, AC-005	UAT-004
BR-001	FR-004	US-003	AC-006, AC-007, AC-008	UAT-005, UAT-006
BR-001	FR-005	US-004	—	UAT-007
BR-001	FR-006	US-005	—	UAT-008
BR-001	FR-007	US-006	AC-009, AC-010, AC-011	UAT-009, UAT-010
BR-005	FR-008	US-007	AC-012, AC-013	UAT-011
BR-001	FR-009	US-008	AC-014, AC-015	UAT-012
BR-002	FR-010	US-012	AC-018	UAT-013
BR-004	FR-011	US-005, US-006	—	UAT-014
BR-005	FR-012	US-006	—	UAT-015
BR-006	FR-013	US-009	AC-016, AC-017	UAT-016
BR-006	FR-014	US-010	—	UAT-017
BR-006	FR-015	US-011	—	UAT-018
BR-007	FR-016	US-014	—	UAT-019
BR-008	FR-017	US-013	—	UAT-020
BR-009	FR-018	US-016	—	UAT-021
BR-010	NFR-003	US-022	AC-021	UAT-022
BR-011	NFR-005, NFR-006	—	—	UAT-023

20. Edge Cases

ID	Edge Case	Scenario	Expected System Behaviour
EC-001	Duplicate request	Customer clicks "Confirm" twice rapidly	System processes first request; second rejected with message "This action has already been processed."
EC-002	Invalid input	Customer enters invalid postal code	System displays validation error; prevents submission.
EC-003	Missing data	Order has no shipping address on file	System displays error: "Unable to process request. Please contact support."
EC-004	Concurrent actions	Two sessions modify same order simultaneously	First action processed; second rejected with "Order has been modified. Please refresh."
EC-005	Service unavailable	Eligibility service is down	System displays: "We are unable to process your request at this time. Please try again later."
EC-006	Timeout	OMS update takes > 30 seconds	System retries; if still fails, displays error and logs issue.
EC-007	Permission issue	Customer attempts to modify another customer's order	System denies access; logs security event.
EC-008	Partial failure	OMS update succeeds but payment refund fails	System logs partial failure; notifies finance team; customer sees "Refund pending" status.
EC-009	Stale data	Order status changed between eligibility check and confirmation	System re-validates before processing; if ineligible, displays updated reason.
EC-010	Conflicting updates	Customer modifies address while warehouse updates shipping status	System uses latest status; if ineligible, rejects modification.
EC-011	Integration failure	WMS is unreachable	System queues inventory release; retries; notifies operations if persistent.
EC-012	Notification failure	Email service is down	System logs failure; queues retry; customer still sees success in portal.
EC-013	Order already shipped	Customer attempts to cancel after shipment	System displays: "This order has already shipped and cannot be cancelled."
EC-014	Partial order eligibility	Some items eligible, some not	System allows cancellation of eligible items only; displays which items cannot be cancelled.
EC-015	Refund amount exceeds original	System calculates refund incorrectly	System validates refund amount against original payment; prevents over-refund.
EC-016	Discount code used	Customer cancels order with discount code	System cancels order; discount code not reissued (per policy); displays message.
EC-017	Gift card payment	Order paid partially with gift card	System refunds gift card portion first, then credit card.
EC-018	Multiple payment methods	Order paid with split payment	System refunds to original payment methods proportionally.
EC-019	Address outside service area	Customer enters address in unserviceable area	System displays: "We do not currently deliver to this address."
EC-020	Shipping speed unavailable	Customer selects express but express unavailable for address	System displays: "Express shipping is not available for this address."
EC-021	Order in picking stage	Customer attempts modification while order is being picked	System displays: "This order is being prepared and cannot be modified."
EC-022	System maintenance	Customer attempts action during maintenance window	System displays: "We are performing maintenance. Please try again later."
EC-023	Browser back button	Customer navigates back after confirming	System displays order details with updated status; no duplicate processing.
EC-024	Session timeout	Customer's session expires during modification	System redirects to login; after login, customer can retry.
EC-025	Currency mismatch	Order in USD; refund processed in different currency	System handles currency conversion; displays refund in original currency.

21. Data / Validation Requirements

21.1 Data Requirements

ID	Data Element	Source	Format	Validation
DR-001	Order ID	OMS	Alphanumeric	Must exist in OMS
DR-002	Order status	OMS	Enum	Must be valid status
DR-003	Order items	OMS	Array	Must have at least 1 item
DR-004	Shipping address	OMS	Structured	Must be valid address format
DR-005	Shipping speed	OMS	Enum	Must be Standard or Express
DR-006	Payment method	Payment Gateway	Tokenized	Must be valid token
DR-007	Refund amount	Calculated	Decimal	Must be ≥ 0 and $\leq$ original amount
DR-008	Customer ID	Auth Service	UUID	Must match logged-in user
DR-009	Timestamp	System	ISO 8601	Must be current
DR-010	Action type	System	Enum	Must be valid action

21.2 Validation Rules

ID	Field	Rule	Error Message
VR-001	Postal code	Must be valid US ZIP code format	"Please enter a valid ZIP code."
VR-002	Address line 1	Required; max 100 characters	"Address is required."
VR-003	City	Required; max 50 characters	"City is required."
VR-004	State	Required; must be valid US state	"Please select a valid state."
VR-005	Shipping speed	Must be available for address	"This shipping speed is not available for your address."
VR-006	Refund amount	Must not exceed original payment	"Refund amount exceeds original payment."
VR-007	Cancellation reason	Optional; max 200 characters	—
VR-008	Confirmation	Must be explicitly checked/clicked	"Please confirm your action."

22. UX/UI Requirements

Screen 1: Order Details Page (Modified)

Element	Description
Purpose	Display order information and available actions
Key Fields	Order ID, order date, items, total, shipping address, shipping speed, order status
Actions	"Modify Order" button, "Cancel Order" button (conditional)
Validation	Buttons only shown if eligible
Error States	"Unable to load order details" if OMS unavailable
Success States	Order details displayed
Accessibility	Screen reader compatible; keyboard navigable

Screen 2: Modification Options Page

Element	Description
Purpose	Display available modification options
Key Fields	List of options: Modify Address, Change Shipping Speed, Cancel Items
Actions	Select option
Validation	Options only shown if eligible
Error States	"No modifications available" if none
Success States	Options displayed
Accessibility	Clear labels; focus management

Screen 3: Address Modification Form

Element	Description
Purpose	Allow customer to enter new address
Key Fields	Address line 1, Address line 2, City, State, ZIP
Actions	"Save Address," "Cancel"
Validation	Required fields; valid ZIP; serviceable area
Error States	Field-level validation errors
Success States	Address saved; confirmation displayed
Accessibility	Labels associated with fields; error announcements

Screen 4: Cancellation Confirmation

Element	Description
Purpose	Confirm cancellation and display refund
Key Fields	Order summary, refund amount, fees, cancellation reason (optional)
Actions	"Confirm Cancellation," "Go Back"
Validation	Confirmation required
Error States	"Unable to process cancellation"
Success States	Cancellation confirmed; success message
Accessibility	Clear confirmation; focus on confirm button

Screen 5: Success Confirmation

Element	Description
Purpose	Confirm successful action
Key Fields	Action summary, updated order status, refund amount (if applicable)
Actions	"View Order," "Return to Home"
Validation	None
Error States	None
Success States	Success message displayed
Accessibility	Clear success message; screen reader announcement

23. API / Integration Considerations

23.1 API Inventory (Conceptual)

API ID	Name	Purpose	Direction	Data Exchanged	Failure Handling
API-001	Order Status API	Retrieve real-time order status	Outbound (to OMS)	Order ID, status, items, fulfillment stage	Retry 3x; fallback to cached data; display error
API-002	Eligibility API	Evaluate modification/cancellation eligibility	Internal	Order data, business rules	Retry 2x; display "try again" message
API-003	Order Update API	Update order in OMS	Outbound (to OMS)	Order ID, modification details	Retry 3x; log failure; notify operations
API-004	Inventory Release API	Release inventory in WMS	Outbound (to WMS)	Order ID, item IDs	Queue retry; notify operations if persistent
API-005	Refund API	Process refund via payment gateway	Outbound (to Stripe)	Payment token, amount, reason	Retry 3x; log failure; notify finance
API-006	Notification API	Send email/SMS/push	Outbound (to providers)	Recipient, template, data	Queue retry; log failure
API-007	Audit Log API	Write audit log entry	Internal	User ID, action, timestamp, before/after	Async; log locally if fails

23.2 Integration Failure Scenarios

ID	Failure Scenario	Impact	Handling
INT-001	OMS API unavailable	Cannot evaluate eligibility or update order	Display error; retry; fallback to support contact
INT-002	WMS API unavailable	Inventory not released	Queue release; retry; notify operations
INT-003	Payment gateway unavailable	Refund not processed	Queue refund; retry; notify finance; customer sees "Refund pending"
INT-004	Notification service unavailable	Customer not notified	Queue notification; retry; log failure
INT-005	Partial failure (OMS updated, refund failed)	Order updated but refund pending	Log partial failure; notify finance; customer sees mixed status
INT-006	Timeout on any API	Operation incomplete	Retry with exponential backoff; if persistent, log and notify

24. Error Handling

ID	Error Scenario	Code	User Message	System Action
ERR-001	Order not found	404	"Order not found. Please check your order number."	Log error; display message
ERR-002	Order not eligible	403	"This order cannot be modified. [Reason]"	Log; display reason
ERR-003	Validation error	400	"Please correct the highlighted fields."	Highlight fields; prevent submission
ERR-004	Service unavailable	503	"We are unable to process your request at this time. Please try again later."	Log; display message; retry option
ERR-005	Timeout	504	"The request timed out. Please try again."	Log; display message; retry option
ERR-006	Payment failure	402	"We were unable to process your refund. Our team has been notified."	Log; notify finance; display message
ERR-007	Inventory release failure	500	"Your cancellation was processed, but inventory update is pending."	Log; queue retry; notify operations
ERR-008	Notification failure	500	(No user-facing error; logged only)	Log; queue retry
ERR-009	Concurrent modification	409	"This order has been modified. Please refresh and try again."	Log; display message
ERR-010	Session expired	401	"Your session has expired. Please log in again."	Redirect to login
ERR-011	Permission denied	403	"You do not have permission to modify this order."	Log security event; display message
ERR-012	System maintenance	503	"We are performing maintenance. Please try again later."	Display maintenance message

25. Testing / UAT Scenarios

Test ID	Req ID	Scenario	Preconditions	Test Steps	Expected Result	Priority
UAT-001	FR-001, FR-002	Eligible order displays modification options	Order is in "Processing" status; customer logged in	1. Navigate to order details. 2. Observe buttons.	"Modify Order" and "Cancel Order" buttons displayed.	High
UAT-002	FR-003	Ineligible order displays reason	Order is "Shipped"; customer logged in	1. Navigate to order details. 2. Attempt to modify.	Message displayed: "This order has already shipped and cannot be modified."	High
UAT-003	FR-002	Real-time eligibility evaluation	Order is in "Processing" status	1. Initiate modification. 2. Observe system response.	Eligibility evaluated within 3 seconds.	High
UAT-004	FR-003	Ineligibility reason for cancellation	Order is "Shipped"	1. Attempt to cancel. 2. Observe message.	Reason displayed: "This order has already shipped."	High
UAT-005	FR-004	Modify delivery address	Order eligible; customer logged in	1. Select "Modify Address." 2. Enter new valid address. 3. Confirm.	Address updated in OMS; confirmation displayed.	High
UAT-006	FR-004	Invalid address validation	Order eligible	1. Select "Modify Address." 2. Enter invalid ZIP. 3. Submit.	Validation error displayed; submission prevented.	High
UAT-007	FR-005	Change shipping speed	Order eligible	1. Select "Change Shipping Speed." 2. Select Express. 3. Confirm.	Shipping speed updated; price difference processed.	Medium
UAT-008	FR-006	Cancel individual item	Order with multiple items; one eligible	1. Select "Cancel" for eligible item. 2. Confirm.	Item cancelled; refund processed; inventory released.	Medium
UAT-009	FR-007	Cancel entire order	Order eligible	1. Select "Cancel Order." 2. Confirm.	Order cancelled; full refund processed; inventory released.	High
UAT-010	FR-007	Cancel ineligible order	Order "Shipped"	1. Attempt to cancel. 2. Observe.	Cancellation prevented; reason displayed.	High
UAT-011	FR-008	Display refund amount	Order eligible	1. Select "Cancel Order." 2. Observe refund display.	Refund amount displayed correctly.	High
UAT-012	FR-009	Confirmation required	Order eligible	1. Select "Cancel Order." 2. Do not confirm.	No action taken until confirmation.	High
UAT-013	FR-010	OMS update	Order eligible	1. Cancel order. 2. Check OMS.	OMS updated with cancellation.	High
UAT-014	FR-011	Inventory release	Order eligible	1. Cancel order. 2. Check WMS.	Inventory released for cancelled items.	High
UAT-015	FR-012	Refund processing	Order eligible	1. Cancel order. 2. Check payment gateway.	Refund initiated.	High
UAT-016	FR-013	Email confirmation	Order eligible	1. Cancel order. 2. Check email.	Confirmation email received within 5 minutes.	High
UAT-017	FR-014	SMS notification	Customer opted in	1. Cancel order. 2. Check SMS.	SMS received.	Medium

Test ID	Req ID	Scenario	Preconditions	Test Steps	Expected Result	Priority
UAT-018	FR-015	Push notification	App installed	1. Cancel order. 2. Check push.	Push notification received.	Medium
UAT-019	FR-016	Audit log	Order eligible	1. Cancel order. 2. Check audit log.	Action logged with timestamp, user, action, before/after.	Medium
UAT-020	FR-017	Support agent view	Agent logged in	1. Open customer record. 2. View self-service actions.	Actions visible.	Medium
UAT-021	FR-018	Operational report	Data available	1. Generate report. 2. View metrics.	Report generated with accurate data.	Medium
UAT-022	NFR-003	Concurrent requests	500 concurrent users	1. Simulate 500 concurrent requests. 2. Monitor performance.	System handles without degradation.	Medium
UAT-023	NFR-005, NFR-006	Security compliance	System deployed	1. Verify TLS. 2. Verify encryption at rest. 3. Verify PCI compliance.	All security requirements met.	High
UAT-024	EC-001	Duplicate request	Order eligible	1. Click "Confirm" twice rapidly.	First request processed; second rejected with message.	High
UAT-025	EC-005	Service unavailable	Eligibility service down	1. Attempt modification. 2. Observe.	Error message displayed; retry option available.	High
UAT-026	EC-008	Partial failure	Refund fails after OMS update	1. Cancel order. 2. Simulate refund failure.	Order cancelled; refund pending; finance notified.	High
UAT-027	EC-009	Stale data	Order status changes during modification	1. Initiate modification. 2. Change order status externally. 3. Confirm.	System re-validates; rejects if ineligible.	High
UAT-028	EC-014	Partial order eligibility	Some items eligible, some not	1. Attempt to cancel ineligible item.	Only eligible items can be cancelled; message displayed.	High
UAT-029	EC-021	Order in picking stage	Order being picked	1. Attempt to modify.	Modification prevented; message displayed.	High
UAT-030	EC-024	Session timeout	Session expires during modification	1. Start modification. 2. Wait for session timeout. 3. Attempt to confirm.	Redirected to login; after login, can retry.	Medium

26. Risk Register

ID	Risk	Likelihood	Impact	Score	Mitigation	Owner
RISK-001	OMS API does not support real-time queries	Medium	High	High	Technical spike early; fallback to cached data	Engineering Lead
RISK-002	WMS integration is complex	Medium	High	High	Engage WMS vendor early; phased rollout	Engineering Lead
RISK-003	Business rules are more complex than anticipated	High	Medium	High	Thorough requirements workshops; iterative rule definition	Product Manager
RISK-004	Customers do not adopt self-service	Medium	Medium	Medium	UX testing; marketing; incentives	Product Manager
RISK-005	Support team resistance to change	Medium	Medium	Medium	Change management; training; clear communication	Support Manager
RISK-006	Peak volume exceeds capacity	Low	High	Medium	Load testing; auto-scaling; rate limiting	Engineering Lead
RISK-007	Refund errors cause financial discrepancies	Low	High	Medium	Automated validation; finance review; audit trail	Finance Manager
RISK-008	Compliance requirements change	Low	Medium	Low	Regular legal review; flexible design	Legal
RISK-009	Integration failures cause poor customer experience	Medium	High	High	Graceful degradation; retry logic; clear error messages	Engineering Lead
RISK-010	Scope creep delays launch	High	Medium	High	Strict scope control; MoSCoW prioritization	Product Manager

27. Assumptions

ID	Assumption	Impact if Invalid	Validation Approach
AS-001	OMS provides real-time order status via API	Eligibility checks would be delayed or inaccurate	Technical spike; API contract review
AS-002	WMS can accept inventory release commands via API	Manual inventory release required	Technical spike; WMS vendor confirmation
AS-003	Payment gateway supports partial refunds	Full refunds only; reduced flexibility	Stripe documentation review
AS-004	Customers have access to web or mobile app	Some customers cannot self-serve	Analytics review; customer survey
AS-005	Business rules for eligibility are stable	Frequent rule changes require agile configuration	Stakeholder interviews; rule documentation
AS-006	Support team will adopt new exception-handling process	Support may continue handling all requests	Training plan; change management
AS-007	Peak volume is 3x normal	System may not handle unexpected spikes	Load testing; capacity planning
AS-008	Email/SMS delivery rates are acceptable	Customers may not receive notifications	Vendor SLA review
AS-009	Legal/Compliance requirements are understood	Compliance risk	Legal review
AS-010	No major changes to order lifecycle during implementation	Requirements may become outdated	Regular stakeholder check-ins

28. Dependencies

ID	Dependency	Type	Owner	Risk if Unavailable
DEP-001	OMS API for real-time order status	Technical	Engineering	Cannot evaluate eligibility
DEP-002	WMS API for inventory release	Technical	Engineering	Inventory not released automatically
DEP-003	Stripe API for refunds	Technical	Engineering	Refunds not automated
DEP-004	Notification service (email/SMS/push)	Technical	Engineering	Customers not notified
DEP-005	UX design for portal/app interfaces	Design	UX Designer	Cannot build UI
DEP-006	Business rules documentation	Business	Product Manager	Cannot codify eligibility
DEP-007	QA environment with test data	Testing	QA Lead	Cannot test
DEP-008	Support team training	Operational	Support Manager	Poor exception handling
DEP-009	Legal/Compliance sign-off	Compliance	Legal	Cannot launch
DEP-010	Analytics instrumentation	Data	Data Team	Cannot measure KPIs

29. Success Metrics / KPIs

ID	KPI	Definition	Baseline (Sim.)	Target (Proposed)	Measurement
KPI-001	Self-Service Adoption Rate	% of eligible modification/cancellation requests handled via self-service	0%	60%	Analytics tracking
KPI-002	Support Contact Reduction	% reduction in support contacts related to order changes	0%	40%	CRM data
KPI-003	Customer Satisfaction (CSAT)	Average CSAT for order change experience	3.2/5	4.2/5	Post-interaction survey
KPI-004	Processing Time	Average time to complete modification/cancellation	10 minutes	< 2 minutes	System logs
KPI-005	Unnecessary Shipments	% of orders shipped despite cancellation request	8%	4%	OMS/WMS data
KPI-006	Refund Processing Time	Average time for refund to be initiated	24+ hours	< 1 hour	Payment gateway data
KPI-007	Feature Error Rate	% of self-service requests resulting in errors	—	< 2%	System logs
KPI-008	System Availability	% uptime for feature	—	99.9%	Monitoring
KPI-009	Notification Delivery Rate	% of notifications successfully delivered	—	> 98%	Notification service logs
KPI-010	Support Agent Efficiency	Average handling time for exception cases	—	< 5 minutes	CRM data
KPI-011	Customer Retention	% of customers who make another purchase after using feature	—	> 70%	Analytics
KPI-012	Revenue Impact	Net revenue impact from reduced cancellations	—	Positive	Financial analysis

30. Implementation / Delivery Roadmap

Phase	Duration	Focus	Key Deliverables
Phase 0: Discovery	2 weeks	Requirements validation; technical spike	Validated requirements; technical feasibility report
Phase 1: Design	3 weeks	UX/UI design; API design; rules engine design	Wireframes; API contracts; rules documentation
Phase 2: Development (MVP)	8 weeks	Core feature development	Eligibility engine; order orchestration; portal UI; payment integration
Phase 3: Testing	3 weeks	QA; UAT	Test cases; UAT sign-off; bug fixes
Phase 4: Pilot	2 weeks	Limited release (10% users)	Pilot metrics; feedback; adjustments
Phase 5: Full Launch	1 week	Full release	Production deployment; monitoring
Phase 6: Phase 2 Features	6 weeks	Should-have features	SMS/push; audit trail; reporting; support dashboard
Phase 7: Optimization	Ongoing	Continuous improvement	KPI monitoring; iterative improvements

30.1 Milestones

Milestone	Target Date	Criteria
M1: Requirements Sign-off	Week 2	Stakeholder approval of requirements
M2: Design Complete	Week 5	UX/UI and API designs approved
M3: MVP Development Complete	Week 13	Core feature functional

Milestone	Target Date	Criteria
M4: Testing Complete	Week 16	All critical bugs resolved
M5: Pilot Launch	Week 18	10% user rollout
M6: Full Launch	Week 19	100% user rollout
M7: Phase 2 Complete	Week 25	Should-have features deployed

31. Expected Business Impact

31.1 Expected / Proposed Impact (Simulated)

Impact Area	Expected Outcome	Rationale
Operational Efficiency	40% reduction in support contacts for order changes	Self-service handles routine requests
Cost Savings	~\$7,200/month support cost reduction	Reduced handling time and volume
Customer Experience	CSAT improvement from 3.2 to 4.2	Faster, more convenient process
Revenue Protection	15% reduction in order abandonment	Customers can modify instead of cancel
Inventory Efficiency	50% reduction in unnecessary shipments	Faster cancellation processing
Scalability	Support volume decoupled from order volume	Self-service scales automatically

31.2 Cost-Benefit Analysis (Simulated)

Item	Estimated Cost/Value
Development Cost	~\$120,000 (simulated)
Infrastructure Cost	~\$2,000/month (simulated)
Maintenance Cost	~\$3,000/month (simulated)
Monthly Support Savings	~\$7,200 (simulated)
Monthly Revenue Protection	~\$15,000 (simulated)
Payback Period	~8 months (simulated)
3-Year ROI	~280% (simulated)

31.3 Actual Measured Results

None. This is a simulated portfolio project. No actual implementation has occurred, and no real data has been measured.

32. AI-Assisted BA Workflow

Activity	AI Role	Human Validation
Brainstorming	Generated feature ideas; suggested business contexts	Selected and refined ideas based on portfolio goals
Requirement Drafting	Drafted initial BR, FR, NFR statements	Reviewed for clarity, specificity, and traceability
Edge Case Discovery	Generated comprehensive list of edge cases	Validated relevance and added domain-specific cases
Requirement Critique	Identified vague or ambiguous requirements	Refined requirements based on feedback
Test Scenario Generation	Drafted UAT scenarios from requirements	Validated coverage and traceability
Documentation Structure	Suggested project structure and section organization	Adapted for portfolio presentation
Consistency Checking	Verified ID consistency across sections	Final human review

AI is an assistant, not the final decision-maker. Human validation remains part of the workflow.

33. Limitations

ID	Limitation	Impact
LIM-001	Simulated company and data	No real-world validation of assumptions

ID	Limitation	Impact
LIM-002	No actual stakeholder interviews	Requirements based on hypothetical needs
LIM-003	No production system access	Technical feasibility not fully validated
LIM-004	No real user testing	UX assumptions not validated
LIM-005	No actual metrics	KPIs are proposed, not measured
LIM-006	Limited technical depth	API specifications are conceptual, not production-ready
LIM-007	No legal review	Compliance requirements are assumed
LIM-008	Single BA perspective	No team collaboration represented
LIM-009	Static documentation	No iterative refinement based on feedback
LIM-010	Portfolio context	Some simplifications for presentation clarity

34. Conclusion

This project demonstrates a complete IT Business Analyst workflow for specifying a software feature from business problem through to UAT scenarios. The Intelligent Order Modification & Cancellation System for the fictional company UrbanThread addresses a realistic business problem with a well-structured, traceable, and testable set of requirements. It demonstrates business analysis skills (problem definition, root cause analysis, stakeholder analysis, AS-IS/TO-BE modeling), IT BA skills (functional and non-functional requirements, user stories, acceptance criteria, use cases, edge cases, API/integration considerations, error handling, UAT, traceability), and professional documentation practice. This is a portfolio project in a simulated context and does not claim real client work, real stakeholder interviews, production data analysis, measured business results, or real development team collaboration.

Artifact Index

The following companion artifacts accompany this case study:

Artifact	Contents
Requirements-Workbook.xlsx	14 sheets: Executive Dashboard, Stakeholder Matrix, Business Requirements, Functional Requirements, Non-Functional Requirements, User Stories, Acceptance Criteria, Use Cases, MoSCoW Prioritization, Traceability Matrix, UAT Scenarios, Risk Register, Assumptions & Dependencies, KPI Success Metrics.
Diagrams/01-AS-IS-Process-Flow.png	Swimlane view of the current manual process.
Diagrams/02-TO-BE-Process-Flow.png	Swimlane view of the proposed self-service process.
Diagrams/03-System-Architecture.png	Conceptual system components, integration layer, and services.
Diagrams/04-Feature-Sequence.png	Sequence-style interaction flow for a modification/cancellation request.
Diagrams/05-Implementation-Roadmap.png	Phased roadmap from discovery through Phase 2 features.
README.md	Project overview, navigation guide, and simulated-data disclaimer.